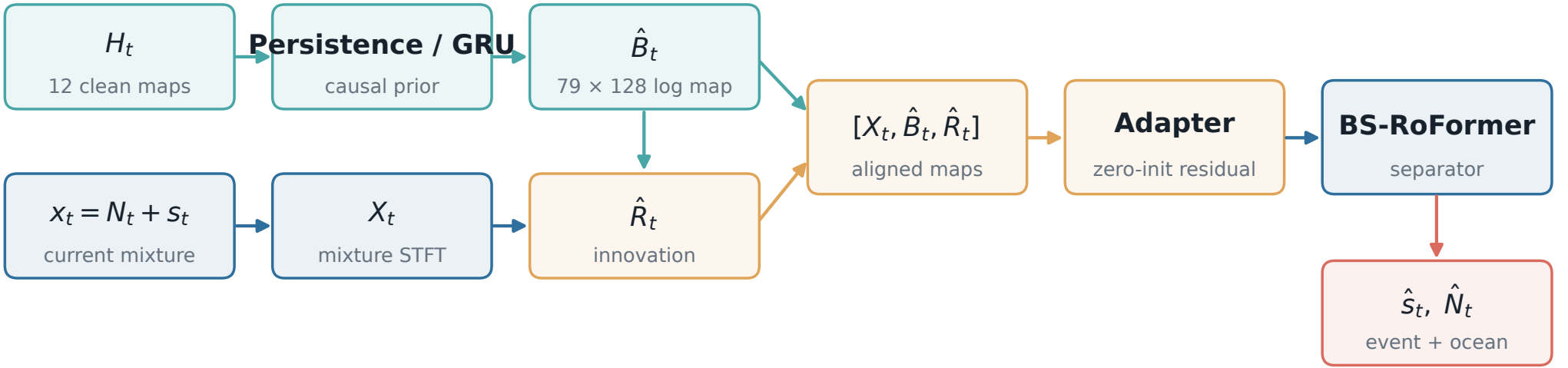
## PRIMARY METHOD · 60-s event-masked history → next 5-s mixture

128 equal-width frequency groups × 79 frames; all history ends strictly before the target window




TRAINING: real clean $N_t$ + synthetic $s_t$ give exact stems; only $x_t$ and past history reach causal arms.

CONTROLS: fixed-mean Zero · other-day Shuffled · last-map Persistence · GRU Forecast · current-background Oracle.